

	DOCUMENT TITLE: Nonconforming Material Report	Revision: A
	DOCUMENT #: FM1-QM.SLQ040	Effective Date: 25Sep2020

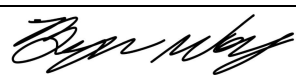
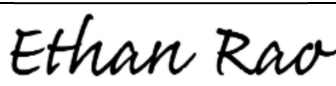
Nonconforming Material Report (NCMR)

NCMR#:	NCMR-0001
Section A - Identification	
Initiated By / Date:	Bryan Wong / 14 Mar 2025
Part Number / Revision:	C.SLQ001 Rev. C
Part or Material Description:	Suspension
Area of Discovery:	QC testing of suspension per QC-C.SLQ001
Batch or Lot Number(s):	2502-01-1
Supplier(s):	Silq
Description of Non-Conformance: <i>(attach supporting documentation, if applicable)</i>	The average suspension concentration of the samples was 8.55 g/L which is less than the target > 10 g/L.
Required Segregation Actions Completed:	☒ Yes ☐ No ☐ Other:
Section B - Evaluation	
Evaluation Performed By / Date:	Ethan Rao / 14 Mar 2025
Evaluation / Root Cause Investigation:	Although not explicitly stated in the manufacturing procedure (MP-C.SLQ001), traveler (TR-C.SLQ001), or quality control procedure or report template (QC-C.SLQ001 and TMP1-QC-C.SLQ001, respectively), the average suspension sample concentration should be greater than 10 g/L in order to complete the subsequent dilution process.

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Section C - Disposition	
Dispositioned By / Date:	Bryan Wong / 17 Mar 2025
Disposition Status:	☐ C ☐ U ☐ RD ☒ RW ☐ RS ☐ S
Temporary Deviation DCO Required:	☐ Yes ☒ No DCO#: N/A
Disposition Rationale:	If the average suspension sample concentration is below 10 g/L, the suspension may be further processed to achieve the desired concentration level. Additional processing would increase the average suspension sample concentration level and would not impact any other characteristic of the suspension. All other acceptance criteria attributes will be measured both before and after the additional processing to ensure there is no effect on the chemical composition of the suspension aside from the concentration. Corrective action: Since additional processing of the suspension can be expected, the manufacturing procedure (MP-C.SLQ001) and traveler (TR-C.SLQ001) will be updated to include the steps to process the suspension further. Additionally, the quality control procedure (QC-C.SLQ001) and report template (TMP1-QC-C.SLQ001) will be updated to include recording of the average suspension sample concentration level.
Rework Instructions:	See attached redlines of the manufacturing procedure, lot traveler, suspension analysis quality control procedure, and quality control report template.
Disposition Approval / Date:	 17 Mar 2025
Disposition Actions Completed:	☐ Scrapped ☒ Reworked ☐ Returned to Supplier ☐ R&D Only Labeled ☐ N/A
Re-test and/or Re-inspection Results: <i>Attach results if applicable</i>	☒ Pass ☐ Fail ☐ N/A Lot released on 29 Apr 2025 

29 Apr 2025

Approval Signatures	
Manufacturing / Operations: 	Date: 17 Mar 2025
Quality Assurance: 	Date: 17 Mar 2025
Engineering: 	Date: 17 Mar 2025



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Suspension Processing

Manufacturing Procedure



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1 Purpose

- 1.1 The purpose of this manufacturing procedure is to provide instructions for the processing of the unpurified polymeric dispersion solution CHEMPSBTCI010 (M.SLQ001) into the Suspension component (C.SLQ001) using the KrosFlow® KTF-2000 Diafiltration System.

2 Scope

- 2.1 This procedure is applicable to the production of the Suspension component (C.SLQ001) for Silq Technologies. The suspension is used to apply the Silq surface treatment (coating) to catheters.

3 Documents

3.1 Associated Documents / Forms

3.1.1 QC-C.SLQ001 Quality Control Procedure, Suspension Analysis

3.1.2 TR-C.SLQ001 Suspension Processing Traveler

3.1.3 QM.SLQ049 Workstation Practices SOP

3.1.3.1.4 QM.SLQ051 Environmental Monitoring SOP

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4 Definitions

4.1 Abbreviations / Acronyms

4.1.1 **COA** – Certificate of Analysis

4.1.2 **DFV** – Designated Fill Volume

4.1.3 **DOV** – Designated Operating Volume

4.2 General Definitions

4.2.1 N/A

5 Responsibilities

- 5.1 It is the responsibility of Manufacturing/Operations management or their authorized representative to ensure this procedure is adequate and appropriate for fulfilling the purpose of this document.
- 5.2 It is the responsibility of Silq personnel and approved contractor personnel to perform their job duties in compliance with the requirements of this procedure and its associated procedures, as appropriate, including ensuring that any output documentation from this procedure is complete and accurate.

6 Safety Information



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- 6.1 Gloves (latex or nitrile) and safety eye wear are to be worn at all times during the execution of this procedure.
- 6.2 This procedure shall be executed in the designated area under a fume hood.

7 Equipment / Tools / Fixtures

- 7.1 KrosFlo KTF-2000 Diafiltration System (E.SLQ006)
- 7.2 High Purity Water Tank Assembly, 750 gallon (E.SLQ002)
- 7.3 Waste Tank Assembly, 500 gallon (E.SLQ004)
- 7.4 Waste diaphragm pump (E.SLQ003)
- 7.5 Concentrate Reactor (E.SLQ009)
- 7.6 Concentrate Reactor Floor Scale (E.SLQ010)
- 7.7 Large Flow Peristaltic Pump (E.SLQ011)
- 7.8 Suspension Floor Scale (E.SLQ012)
- ~~7.9~~ Pneumatic Drum Stirrer (E.SLQ007)
- ~~7.9~~~~7.10~~ Portable Exchange Deionizer System (E.SLQ001)
- ~~7.10~~~~7.11~~ Dymo LabelWriter 5XL Label Printer with Dymo Connect Software

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8 Components / Materials / Supplies

- 8.1 Polymeric Dispersion Solution, Unpurified, ~~CHEMPSBTG1010~~ (M.SLQ001)
- 8.2 Label Stock, 4" x 6", White (C.SLQ002)
- 8.3 Drum, Plastic, 55 gallon, Closed Top (Product Tank) (C.SLQ003)
- 8.4 High Purity Water (S.SLQ001)
- 8.5 Centrifuge tube, 50 mL, Polypropylene, Flat Cap (S.SLQ002)
- 8.6 Container, 8.5" x 18" x 6" (S.SLQ003)
- 8.7 Bleach (S.SLQ007)
- 8.8 Permanent pen

9 Suspension Processing Equipment and Layout

- 9.1 System Diagram: See Appendix A for system diagram for the suspension processing. The system diagram identifies the equipment, tubing, valves, and fittings required for processing the suspension at the Chico facility.
- 9.2 Image of KTF-2000 System

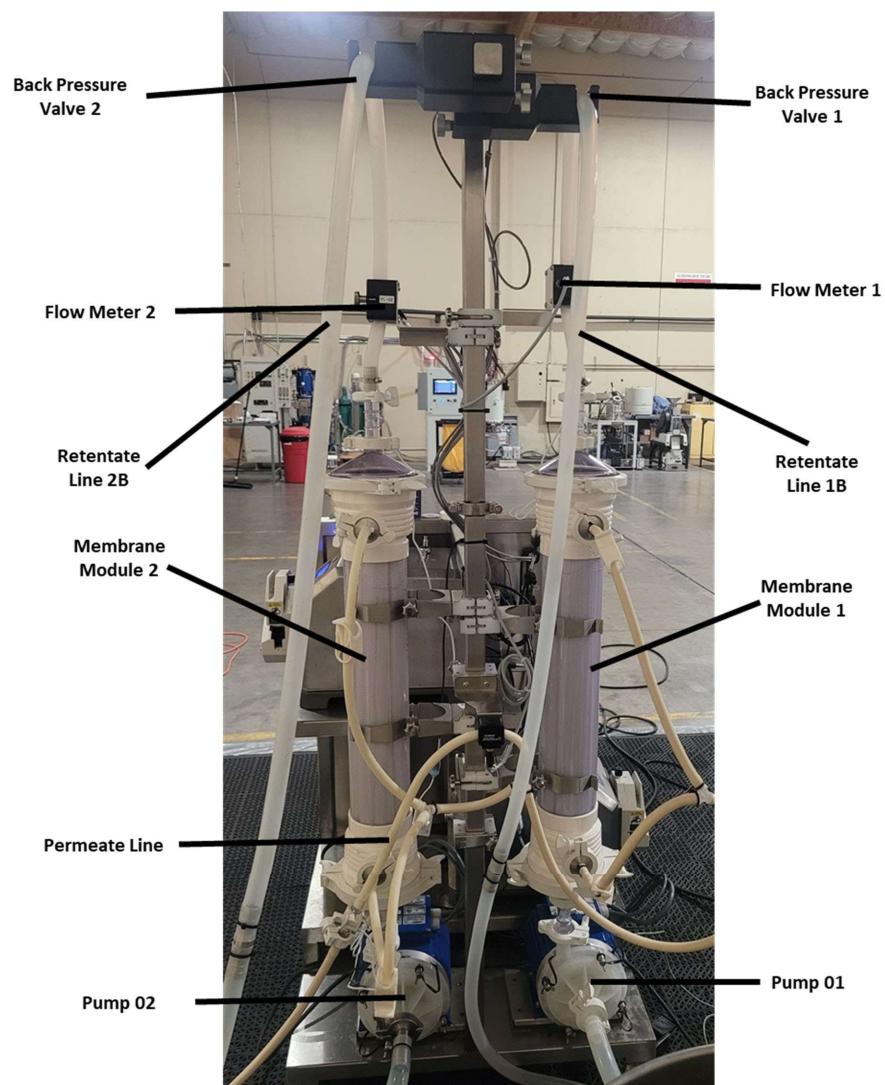


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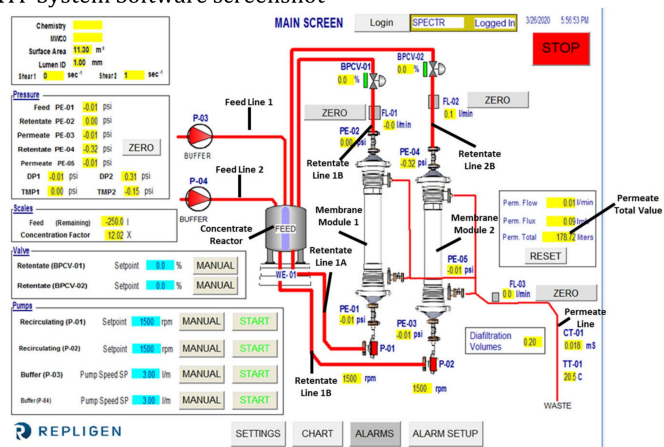
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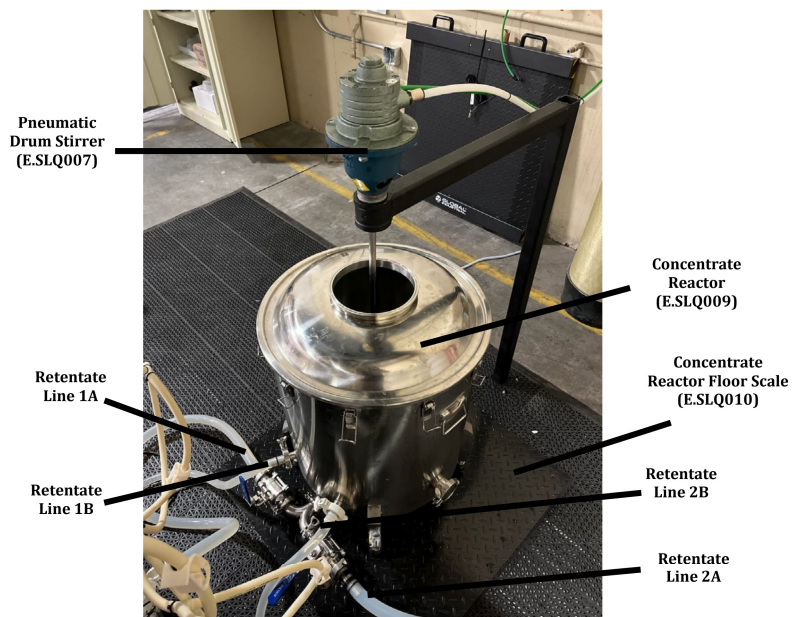
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9.3 KTF System Software screenshot



9.4 Concentrate Reactor, pneumatic drum stirrer and connecting lines





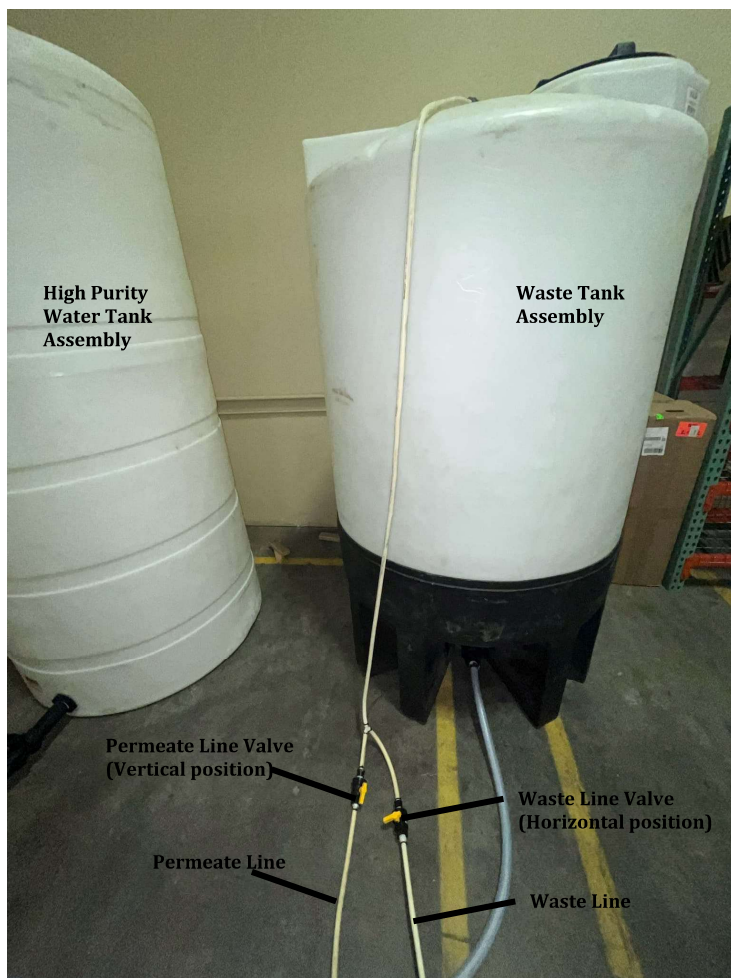
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9.5 High Purity Water Tank Assembly, Waste Tank Assembly, and connecting lines



10 Procedure: Line Clearance

- 10.1 Perform line clearance of the work area in accordance with Workstation Practices QM.SLQ049.



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11 Procedure: KTF-2000 System Initialization

- 11.1 Turn on the System Control Panel of the KTF-2000 System (E.SLQ006).
- 11.2 Initialize system software.
- 11.3 Enter login credentials:
 - 11.3.1 User account: Spectr
 - 11.3.2 Password: 3255.

12 Procedure: System Preparation

- 12.1 Verify the conductivity indicator on the Portable Exchange Deionizer System (E.SLQ001) is green.
- 12.1.1 If the indicator is red, discontinue system preparation contact the appropriate service provider to resolve.
- 12.1.2 System preparation and production may continue at risk provided that the High Purity Water (S.SLQ001) at the outlet of the Portable Exchange Deionizer System is tested for water quality in accordance with QM.SLQ051.
- 12.412.2 Place Containers (S.SLQ003) below Pump 01 and Pump 02 of the KTF-2000 System.
- 12.512.3 Disconnect the clamps holding the plug on Pumps 01 and 02 and allow liquid to drain into the containers.
- 12.312.4 Turn the Waste Line Valve into the vertical (open) position and the Permeate Line Valve into the horizontal (closed) position (figure in Section 9.5 indicates the valve system). Using the Peristaltic Pump (E.SLQ011) and the waste line, drain containers into the waste tank (E.SLQ004).
- 12.412.5 Connect Retentate Line 1A to Pump 01 and Retentate Line 2A to Pump 02.
- 12.512.6 Disconnect the clamps holding the plug above Membrane Module 1 and Membrane Module 2.
- 12.612.7 Connect Retentate Line 1B to Membrane Module 1 and Retentate Line 2B to Membrane Module 2.
- 12.712.8 Place Retentate Line 1B through Flow Meter 1 and Backpressure Valve 1 and Retentate Line 2B through Flow Meter 2 and Backpressure Valve 2.
- 12.812.9 Place Feed Line 1 into Pump 03 and Feed Line 2 into Pump 04.
- 12.912.10 Place Permeate Line into the Peristaltic Pump (E.SLQ011). Change the Waste Line Valve to the horizontal (closed) position and the Permeate Line Valve to the vertical (open) position.

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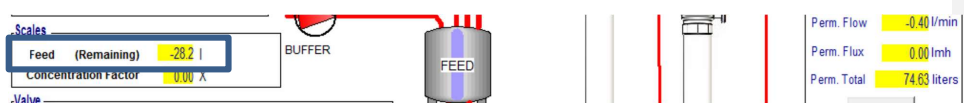
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12.1012.11 Press 'Zero' on Control Panel of the Concentrate Reactor Floor Scale (E.SLQ010) to tare the floor scale.

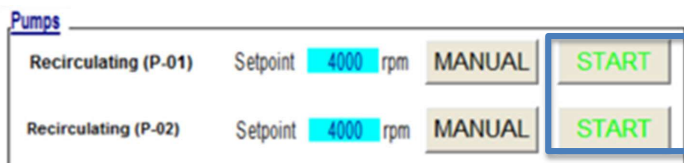
12.1112.12 Fill Concentrate Reactor with High Purity Water (S.SLQ001) until the floor scale reads ~30 kg. Record value in traveler.

12.1212.13 Ensure that the 'Feed (Remaining)' value on the system software is within ± 0.2 kg of the reading on the floor scale.



12.1312.14 Press the START buttons highlighted below on the system software to manually run Pump 01 and Pump 02 as shown below until Membrane Modules 1 and 2 and Retentate Lines 1A, 1B, 2A, and 2B are filled.

Note: User may be required to elevate Retentate Lines 1A or 2A so water is able to reach the pumps.



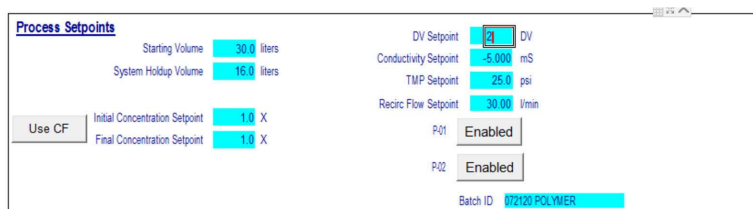
12.1412.15 Ensure all connections are secure and no leaks are present.

12.1512.16 Again, fill Concentrate Reactor until the floor scale reads ~30 kg.

12.1612.17 Set the following parameters on main page of the system software:

12.16.112.17.1 Starting Volume: 30 L

12.16.212.17.2 DV Setpoint: 2.0 DV



12.1712.18 Click the START PROCESS button.



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12.1912.20 Continue to run process until 'Perm. Total' value reads 60L in the system software. Click Stop.



12.2012.21 Mount Pneumatic Drum Stirrer (E.SLQ007) into Holder and turn on air, roughly 45° or more on the air inlet valve so that the water in the Concentrate Reactor is lightly stirred.

13 Procedure: Suspension Processing

- 13.1 Obtain the designated Unpurified Polymeric Dispersion Solution (M.SLQ001) container(s) for the production work order. Record material traceability information on the traveler.
- 13.2 Record volume of container(s) on the traveler.
- 13.3 Fill the Concentrate Reactor with High Purity Water until the Designated Fill Value (DFV) equals the volume of the container(s). Record the DFV in the traveler.

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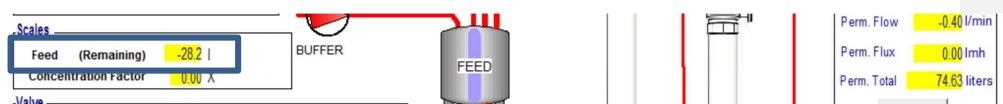
Note: The minimum DFV value and volume of Unpurified Polymeric Dispersion Solution should be 20L.

- 13.4 Calculate the Designated Operating Value (DOV). The DFV is 30% of the DOV.

$$\text{DOV} = 10 / 3 * \text{DFV}$$

Record the DOV in the traveler.

- 13.5 Slowly pour contents of the Unpurified Polymeric Dispersion Solution container(s) into the Concentrate Reactor as the pneumatic drum stirrer continues stirring. Pause if necessary to ensure that no gelation develops.
- 13.6 Fill the Concentrate Reactor with High Purity Water until the DOV is reached (within $\pm 2\text{kg}$) as read by the floor scale and “Feed (Remaining)” field on the system software. Verify both the floor scale and system software are within $\pm 2\text{kg}$ of DOV and each other. Record values on the traveler.



- 13.7 Set parameters on main page of system software as shown

13.7.1 Starting Volume: Designated Operating Value

13.7.2 DV Setpoint: 10 DV



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Process Setpoints

Designated Operating Value

Starting Volume: 30.0 lers
System Holdup Volume: 15.0 lers

Use CF: Initial Concentration Setpoint: 1.0 X
Final Concentration Setpoint: 1.0 X

DV Setpoint: 10.0 DV
Conductivity Setpoint: 2.000 mS
TMP Setpoint: 25.0 psi
Recirc Flow Setpoint: 30.00 l/min

P01: Enabled
P02: Enabled

Batch ID: 0010 POLYMER

Starting Vol. = Enter Above

START PROCESS

STOP

Filter Model

X04.E001-05-N
X04.E001-05-S
X04.E003-05-N
X04.E003-05-S
X04.E003-10-N
X04.E003-10-S
X04.E005-05-N
X04.E005-05-S
X04.E005-10-N
X04.E005-10-S
X04.E010-05-N
X04.E010-05-S

Filter Model: X04.E003-05-N
Fiber Count: 12000
Lumen ID: 0.50 mm
MWCO: 1000
Chemistry: MWC0
Effective Length: 0.0 cm
Surface Area: 12.00 m²
Surface Area Entry: 120000.00 cm²

PID Setup

Recirc Pump (P-01)
P: 3
I: 2
D: 0

Product Pump (P-02)
P: 3
I: 2
D: 0

Buffer Pump (P-03)
P: 10
I: 0
D: 1

Permeate Pump (P-04)
P: 100
I: 0
D: 0

Back Pressure Valve (BPCV-01/02)
Starting Position: 50 % Closed

To RESET control loop windup, Press Below
RESET WINDUP
Assure needed control loops are in MANUAL, before pressing.

RESET PID TO FACTORY SETTINGS

Re-Open Keypad

Tubing Selection

Exit iFix

- 13.8 Click START PROCESS button. Verify both Pump 01 and Pump 02 are running.
Note: User may be again required to elevate Retentate Lines 1A or 2A so solution is able to reach the pumps.



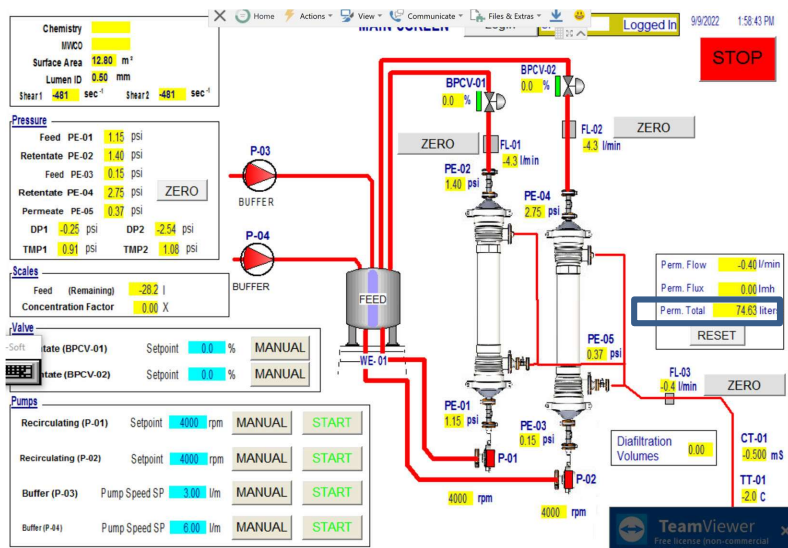
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- 13.9 Run process until 'Perm. Total' value reaches 10 times the value of the DOV. Click STOP button.



14 Procedure: Transferring suspension to final packaging

- 14.1 Place a Product Tank (55 gallon Plastic Drum C.SLQ003) onto the Suspension Floor Scale (E.SLQ012). Record the weight of the product tank on the traveler.
- 14.2 Zero (e.g., tare) the Suspension Floor Scale.
- 14.3 Disconnect the Permeate Line from the Peristaltic Pump and connect Product Line to the Peristaltic Pump (E.SLQ011). Insert the ends of the Product Line into the Concentrate Reactor and Product Tank appropriately. Turn on the Peristaltic Pump to begin the transfer of the contents of the Concentrate Reactor to the Product Tank.
- 14.4 During the suspension transfer process, fill three 50 mL tubes (S.SLQ002) with aliquots of suspension solution for quality analysis at the beginning, middle, and end of the transfer process. Label the tubes Aliquot A, Aliquot B, and Aliquot C, respectively.
- 14.5 Reconnect Permeate Line to the Peristaltic Pump.
- 14.6 Place clean containers (S.SLQ003) below Pump 01 and Pump 02. Disconnect Retentate Line 1A from Pump 01 and Retentate Line 2A from Pump 02. Allow



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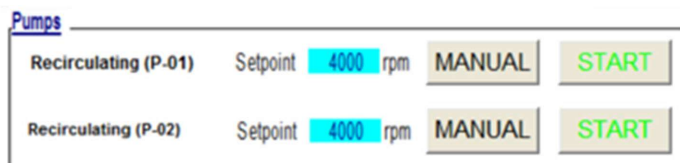
suspension to drain into containers and pour the contents into the Product Tank. Reconnect Retentate Line 1A to Pump 01 and Retentate Line 2A to Pump 02.

- 14.7 Record the Initial Suspension Weight of the Product Tank using Suspension Floor Scale in the traveler.
Assumption: The initial weight (in kg) of the suspension is equal to the initial volume (in L).
- 14.8 Seal the Product Tank.
- 14.9 Print ~~the two copies of the~~ product label for the Suspension (L.SLQ005) entering the ~~part revision~~, lot number, and expiration date. The expiration date is ~~twone~~ (2+) years from the date of manufacturing. Apply ~~the one~~ product label to the product tank. Apply the other product label to the DHR.
- 14.10 Place the product tank in quarantine pending QC analysis of the Suspension and DHR review & release.
- 14.11 Perform QC analysis on suspension samples in accordance with QC-C.SLQ001.
- 14.12 Once the QC analysis has been performed, attach the Suspension Quality Control Report and COA to the DHR and submit to QA for review and release of product.
- 14.13 Confirm test results documented in the COA meet acceptance criteria.

Note: The diafiltration system, equipment, and supplies should be prepared for storage (Section 15) immediately after Step 14.10.

15 Procedure: Prepare diafiltration system, equipment, and supplies for storage

- 15.1 Preparing KTF-2000 System for Storage
- 15.1.1 Fill Concentrate Reactor with High Purity Water until the floor scale reads ~30 kg. Verify that the 'Feed Remaining' matches the reading on the scale.
- 15.1.2 Click the START button to manually run Pump 01 and Pump 02 as shown below until Membrane Modules 1 and 2 and Retentate Lines 1A, 1B, 2A, and 2B are filled.





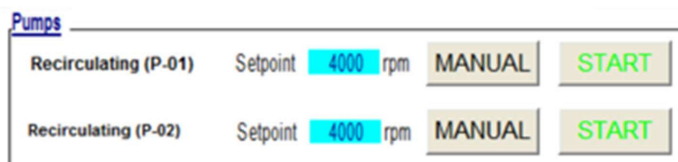
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- 15.1.3 Continue to allow Pump 01 and Pump 02 to run for 10 minutes and then click the STOP button on each pump
- 15.1.4 Switch the Permeate Line Valve to the horizontal (closed) position and the Waste Line Valve to the vertical (open) position.
- 15.1.5 Place Waste Line through the Peristaltic Pump and into the Concentrate Reactor.
- 15.1.6 Manually run the Peristaltic Pump to fully drain the contents of the Concentrate Reactor into the Waste Tank.
- 15.1.7 Add 20L of High Purity Water to clean container followed by 1.5mL of bleach solution (S.SLQ007) to create Storage Solution.
- 15.1.8 Add Storage Solution to Concentrate Reactor. Close the four Permeate Line clamps and switch Permeate Line Valve to horizontal (closed) position and Waste Line Valve to vertical (open) position.
- 15.1.9 Click the START button to manually run Pump 01 and Pump 02 as shown below until Membrane Modules 1 and 2 and Retentate Lines 1A, 1B, 2A, and 2B are filled.



- 15.1.10 Once filled, stop Pump 01 and Pump 02 and close the ball valves for Retentate Lines 1B and 2B at the Concentrate Reactor. Allow 5 minutes to elapse for settling and degassing. Ensure the Storage Solution has filled Membrane Modules 1 and 2 entirely. Storage Solution should at least fill Retentate Lines 1B and 2B up to the respective line's Flow Meter.
- 15.1.11 If the Storage Solution level has fallen below the Flow Meter in either line, repeat step 15.1.9 and 15.1.10.
- 15.1.12 Manually run the Peristaltic Pump to fully drain the contents of the Concentrate Reactor into the Waste Tank.
- 15.1.13 Power down Pump 01, Pump 02, the Concentrate Reactor Scale, Suspension Scale, and the KTF System Software.
- 15.2 Procedure: Preparing the High Purity Water Tank for Storage



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- 15.2.1 The high purity water tank is to be stored empty and dry when not in use.
- 15.2.2 Take a peristaltic pump with sterilized tubing to pump out any residual high purity water from the tank.
- 15.2.3 Allow high purity water tank to dry for 24 hours before securing lid to the tank.
- 15.3 Procedure: Clean supplies
 - 15.3.1 Clean containers used in the manufacturing process with commonly used cleaning agents. Allow containers to dry. Store containers after they have dried.

16 Procedure: Suspension Dilution

The manufacturing process is designed to produce suspension at a concentration greater than the intended target concentration of use (1% w/v). Therefore, after suspension quality analysis has been performed, a specific amount of High Purity Water will be added to the Product Tank to dilute the suspension from its initial concentration to the intended target concentration of use, 1% w/v.

- 16.1 Calculate the Average Suspension Sample Concentration (g/L) by taking the average 'Residue Net Weight' of six (6) freeze dried suspension samples based off the 10 mL sample. (The 'Residue Net Weight' can be found in the Suspension Quality Control Report.)
- 16.2 Calculate the volume (L) of High Purity Water to be added to the product tank using the formula below:
$$\text{Volume} = [\text{Average Suspension Sample Concentration (g/L)} * \text{Initial Suspension Volume (L)}] / [10 \text{ (g/L)}] - \text{Initial Suspension Volume (L)}$$
- 16.3 Record the volume in the traveler.
- 16.4 Obtain the product tank from quarantine. Tare the Suspension Floor Scale.
- 16.5 Place the product tank on the floor scale and add the determined amount of high purity water to the product tank.
- 16.6 Record the weight of the product tank.
- 16.7 Calculate the weight of the suspension by subtracting the weight of the product tank from Step 14.1.
- 16.8 Update the product label for the Suspension (L.SLQ005) with the suspension weight.

17 Suspension Concentration Process



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The manufacturing process is designed to produce suspension at a concentration greater than the intended target concentration of use (1% w/v). If, after suspension quality analysis has been performed, it is found that the polymer concentration of the lot is below the target concentration, this additional concentration step shall be performed and the quality analysis procedure shall be subsequently repeated. Operators shall perform the Suspension Concentration process according to the following steps and fill out the "Build Record Concentration Process" section of the traveler accordingly.

17.1 Calculate the Concentration Permeate Required Value according to the following calculation

17.1.1 Concentration Permeate Required = [12 (g/L)-Average Suspension Sample Concentration (g/L)]/ 12 (g/L) * Initial Suspension Volume (L) + 16 (L)

17.2 Perform System Preparation procedure outlined in Section 12.

17.3 Disconnect the Permeate Line from the Peristaltic Pump and connect Product Line to the Peristaltic Pump (E.SLQ011). Insert the ends of the Product Line into the Concentrate Reactor and Product Tank appropriately. Turn on the Peristaltic Pump to begin the transfer of the contents of the Product Tank to the Concentrate Reactor.

17.4 Run process until 'Perm. Total' value reaches the Concentration Permeate Required Value.

17.5 Perform Procedure: Transferring suspension to final packaging steps outlined in Section 14 including all sampling activities.

17.6 Perform Procedure: Prepare diafiltration system, equipment, and supplies for storage steps outlined in Section 15.



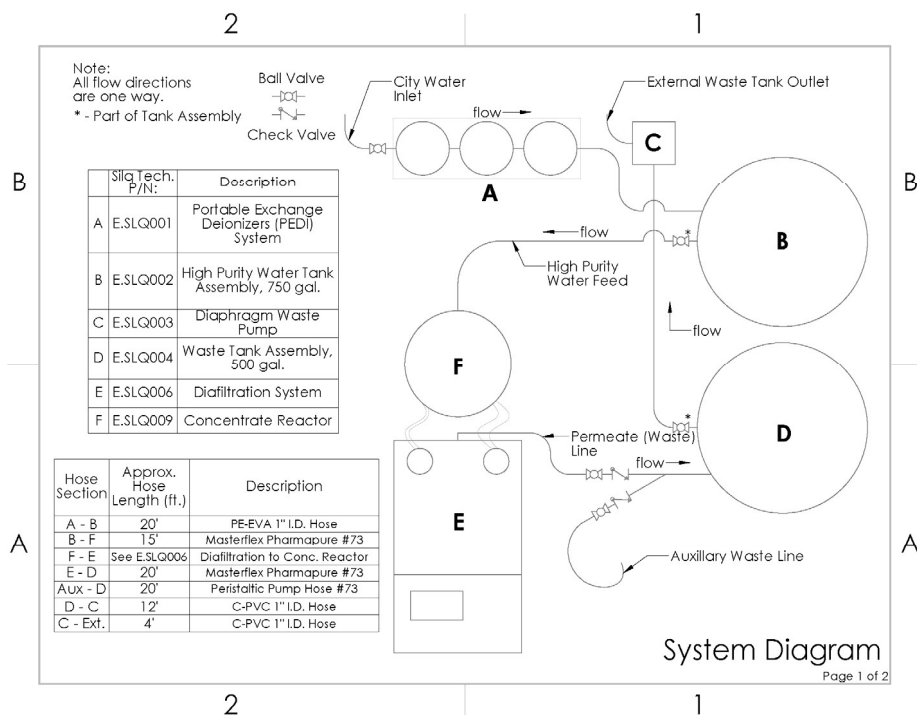
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1718 Appendix A: System Diagram, Suspension Processing





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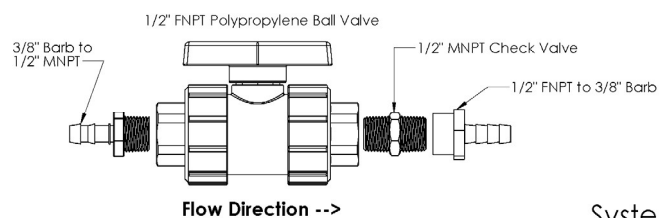
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MP-C.SLQ001

Effective Date:
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Hose Section	Approx. Hose Length (ft.)	Description	Supplier	Part Number
A - B	20'	PE-EVA 1" I.D. Hose	USPLASTICS	59519 BEV-A-LINE IV 1ID X1 1/4ODX 1/8 NAT B45
B - F	15'	Masterflex Pharmapure #73	VWR	MFLX06435-73
F - E	See E.SLQ006	Diafiltration to Concentrate Reactor		
E - D	20'	Masterflex Pharmapure #73	VWR	MFLX06435-73
Aux - D	20'	Persistaltic Pump Hose #73	USPLASTICS	56590 3/8" ID x 1/2" OD x 1/16" Wall Tygon® XL-60 Tubing
D - C	12'	C-PVC 1" I.D. Hose	PBM Supply	CBT10
C - Ext.	4'	C-PVC 1" I.D. Hose	PBM Supply	CBT10

Ball Valve / Check Valve Assembly



System Diagram

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General Production Information

Work Order #:		NCMR #:	
Part Number:	C.SLQ001	Part Revision:	
Lot #:		Manufacture Date / Expiration Date:	Mfg date: Exp date:
Build Purpose:			
Additional Build Info:			

Material Traceability

Part Number	Description	Part Revision	Lot / Batch #	Expiration Date	Qty. Obtained	Obtained By: (Initial/Date)
M.SLQ001	Polymeric Dispersion Solution, Unpurified, CHEMPSBTCI010					

Equipment Traceability

Part Number	Description	Equipment ID	Calibration Due Date
E.SLQ006	Diafiltration System, KrosFlo KTF-2000		
E.SLQ010	Floor Scale, Concentrate Reactor		
E.SLQ012	Floor Scale, Suspension		

Build Record

Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
1	QM.SLQ049	Line clearance completed	☐ Yes ☐ No ☐ N/A	
2	MP-C.SLQ001	Verify conductivity indicator on Portable Exchange Deionizer System (E.SLQ001) is green.	☐ Indicator is green	

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Silq Technologies Corporation	PROPRIETARY AND CONFIDENTIAL This document is for reference only	Page 1 of 11



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Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
32	MP-C.SLQ001	Drain liquid in diafiltration system/tubing into waste tank.	☐ Yes ☐ No	
43	MP-C.SLQ001	Connect Retantate Retentate Lines and Feed Lines to the appropriate pumps. Route lines through the appropriate membrane modules, flow meters, and backpressure valves.	☐ Yes ☐ No	
54	MP-C.SLQ001	Place Permeate Line into Peristaltic Pump; change Valve 1 to 'permeate line' position.	☐ Yes ☐ No	
65	MP-C.SLQ001	Tare Concentrate Reactor Floor Scale. Fill Concentrate Reactor with ~30kg of high purity water (S.SLQ001).	☐ Floor scale tared Floor scale reading: _____	
76	MP-C.SLQ001	'Feed (remaining)' value on system software is within ± 0.2 kg of floor scale reading.	'Feed (Remaining)' value: _____ ☐ Within ± 0.2 kg of floor scale reading	
87	MP-C.SLQ001	Manually run Pump 01 and Pump 02 until membrane modules and retentate lines are filled.	☐ Yes ☐ No	
98	MP-C.SLQ001	Confirm no leaks are observed in all connections.	☐ No leaks observed	
109	MP-C.SLQ001	Fill Concentrate Reactor until scale reads ~30kg.	Floor scale reading: _____	
110	MP-C.SLQ001	Set parameters on main page of system software. - Starting Volume: 30L - DV Setpoint: 2.0 DV	☐ Parameters set	
124	MP-C.SLQ001	Click START PROCESS button to fill membrane modules with water. After membrane modules are filled with water, ensure Permeate Valve is set to vertical position and open the four clamps on permeate line.	☐ Membrane modules filled with water ☐ Permeate Valve set to vertical position ☐ Open four clamps on permeate line	
132	MP-C.SLQ001	Click Stop button when 'Perm. Total' value reads 60L on system software.	'Perm. Total' value: _____	

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Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
143	MP-C.SLQ001	Mount Pneumatic Drum Stirrer into holder and turn on.	☐ Yes ☐ No	
154	MP-C.SLQ001	Obtain container(s) of Unpurified Polymeric Dispersion Solution (M.SLQ001). Record traceability information in the Material Traceability table above.	☐ Container(s) obtained ☐ Material traceability recorded	
165	MP-C.SLQ001	Record volume of container(s).	Volume of container(s): _____	
176	MP-C.SLQ001	Fill Concentrate Reactor with high purity water until the Designated Fill Value (DFV) equals the volume of container(s).	☐ Yes ☐ No Designated Fill Value (DFV): _____	
187	MP-C.SLQ001	Calculate the Designated Operating Value (DOV). $DOV = 10 / 3 * DFV$.	Designated Operating Value (DOV): _____	
198	MP-C.SLQ001	Slowly pour contents of Unpurified Polymeric Dispersion Solution container(s) into concentrate reactor. Ensure no gelation develops.	☐ Yes ☐ No	
2019	MP-C.SLQ001	Fill concentrate reactor with high purity water until DOV is reached (within $\pm 2\text{kg}$). Record floor scale and system software 'Feed (Remaining)' value. Verify values are within $\pm 2\text{kg}$ of DOV and each other.	Floor scale reading: _____ "Feed (Remaining)" value: _____ ☐ Values are within $\pm 2\text{kg}$ of DOV and each other	
210	MP-C.SLQ001	Set parameters on main page of system software. - Starting Volume: Designated Operating Value - DV Setpoint: 10 DV	☐ Parameters set	
221	MP-C.SLQ001	Click START PROCESS button. Verify Pump 01 and Pump 02 are running	☐ Pump 01 and Pump 02 running	
232	MP-C.SLQ001	Run process until "Perm. Total" value reaches 10 times the value of the Designated Fill Value (see Step 17 above).	"Perm. Total" value: _____	

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Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
243	MP-C.SLQ001	Record weight of product tank (C.SLQ003) without the lid.	Product tank weight: _____ kg	
254	MP-C.SLQ001	Transfer Suspension to plastic drum using the peristaltic pump via the product line (C.SLQ003).	☐ Yes ☐ No	
265	MP-C.SLQ001	Take sample of suspension in 50mL tubes at the beginning, middle, and end of the transfer. Label each tube Aliquot A, Aliquot B, and Aliquot C, respectively.	☐ Yes ☐ No	
276	MP-C.SLQ001	Place clean containers below Pump 01 and Pump 02. Disconnect retentate lines from pumps and allow suspension to drain into containers. Pour contents of the containers into the product tank.	☐ Yes ☐ No	
287	MP-C.SLQ001	Record the initial weight of the suspension. Assumption: The initial weight (in kg) of the suspension is equal to the <u>weight of product tank subtracted from the weight of the suspension and product tank initial volume (in L)</u> .	Initial Suspension Weight: _____ kg	
298	MP-C.SLQ001	Seal product tank.	☐ Yes ☐ No	
3029	MP-C.SLQ001	Create product label for Suspension (L.SLQ005) indicating lot number and expiration date of suspension. Apply label to product tank.	☐ Yes ☐ No	
3130	MP-C.SLQ001	Place product tank in quarantine pending QC analysis of suspension.	☐ Yes ☐ No	
321	MP-C.SLQ001	<i>Prepare KTF-2000 System for storage</i> Fill concentrate reactor with ~30kg of high purity water and manually start running Pump 01 and Pump 02. Allow Membrane Modules 1 and 2 and Retentate Lines 1A, 1B, 2A, and 2B are filled. Allow pumps to run for an additional 10 minutes.	☐ Yes ☐ No	

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Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
332	MP-C.SLQ001	Switch Permeate Line Valve to horizontal (closed) position and Waste Line Valve to vertical (open) position. Place Waste Line through peristaltic pump and into the concentrate reactor. Manually run the peristaltic pump to fully drain the contents of the concentrate reactor into the waste tank.	☐ Yes ☐ No	
343	MP-C.SLQ001	Create Storage Solution. Combine 20L of high purity water + 1.5mL of bleach solution (S.SLQ007).	☐ Yes ☐ No	
354	MP-C.SLQ001	Pour Storage Solution into Concentrate Reactor. Close 4 Permeate Line Clamps and switch Permeate Line Valve to horizontal (closed) position and Waste Line Valve to vertical (open) position.	☐ Yes ☐ No	
365	MP-C.SLQ001	Manually run Pump 01 and Pump 02 until Membrane Modules 1 and 2 and Retentate Lines 1A, 1B, 2A, and 2B are filled.	☐ Yes ☐ No	
376	MP-C.SLQ001	Once filled, stop Pump 01 and Pump 02 and close the ball valves for Retentate Lines 1B and 2B at the Concentrate Reactor. Allow 5 minutes to elapse for settling and degassing. Ensure the Storage Solution has filled Membrane Modules 1 and 2 entirely. Storage Solution should at least fill Retentate Lines 1B and 2B up to the respective line's Flow Meter.	☐ Yes ☐ No	
387	MP-C.SLQ001	Wait 5 minutes to ensure the Membrane Modules are full and stable.	☐ Yes ☐ No	
398	MP-C.SLQ001	Manually run the Peristaltic Pump to fully drain the contents of the Concentrate Reactor into the Waste Tank.	☐ Yes ☐ No	
4039	MP-C.SLQ001	Power down Pump 03, Pump 04, concentrate reactor scale, suspension scale, and KTF system software.	☐ Yes ☐ No	
410	MP-C.SLQ001	Prepare the high purity water tank for storage Take a peristaltic pump with sterilized tubing to pump out any residual high purity water from the tank.	☐ Yes ☐ No	
421	MP-C.SLQ001	Allow high purity tank to dry for 24 hours before securing lid to the tank.	☐ Yes ☐ No	

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Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
432	MP-C.SLQ001	Clean containers using in the manufacturing process. Allow containers to dry and store after they have dried.	☐ Yes ☐ No	
443	QC-C.SLQ001	Execution of QC analysis of suspension samples. QC analysis report attached to DHR.	☐ Yes ☐ No	
454	MP-C.SLQ001	Attach Suspension Quality Control Report and COA to DHR.	☐ Yes ☐ No	
465	MP-C.SLQ001	Test results meet acceptance criteria.	☐ Yes ☐ No	
476	MP-C.SLQ001	<i>Dilution Procedure</i> Calculate the Average Suspension Sample Concentration (g/L) by taking the average 'Residue Net Weight' of six (6) freeze dried suspension samples based off the 10 mL sample.	Average Suspension Sample Concentration: _____ g/L	
48	MP-C.SLQ001	<u>Determine if Concentration Process is required. If Average Suspension Sample Concentration is <10g/L, Concentrate Process is required. If Average Suspension Sample Concentration is >10g/L, Dilution Procedure can proceed.</u>	<u>Is Concentration Process Required?</u> ☐ Yes, skip Steps 49-54 and proceed to Step A1. ☐ No, proceed to Step 49.	
497	MP-C.SLQ001	Calculate the volume (L) of high purity water to be added to the product tank. Volume = [Average Suspension Sample Concentration (g/L) * Initial Suspension Volume (L)]/[10 (g/L)] – Initial Suspension Volume (L)	Volume of high purity water to be added to product tank: _____ L	
5048	MP-C.SLQ001	Tare suspension floor scale and place product tank onto floor scale.	☐ Yes ☐ No	
5149	MP-C.SLQ001	Add volume of high purity water (from Step 35) to the product tank.	☐ Yes ☐ No	
5250	MP-C.SLQ001	Record the weight of product tank <u>with suspension</u> .	_____ kg	
534	MP-C.SLQ001	Calculate weight of suspension.	_____ kg	
542	MP-C.SLQ001	Update product label with weight of suspension.	☐ Yes ☐ No	

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Comments/Additional Notes:

Build Record Suspension Concentration Process

(Check one: ☐ Required, ☐ Not Required)

Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
A1	QM.SLQ049	Line clearance completed	☐ Yes ☐ No ☐ N/A	
A2	MP-C.SLQ001	Drain liquid in diafiltration system/tubing into waste tank.	☐ Yes ☐ No	
A3	MP-C.SLQ001	Connect Retentate Lines and Feed Lines to the appropriate pumps. Route lines through the appropriate membrane modules, flow meters, and backpressure valves.	☐ Yes ☐ No	
A4	MP-C.SLQ001	Place Permeate Line into Peristaltic Pump; change Valve 1 to 'permeate line' position.	☐ Yes ☐ No	
A5	MP-C.SLQ001	Tare Concentrate Reactor Floor Scale. Fill Concentrate Reactor with ~30kg of high purity water (S.SLQ001).	☐ Floor scale tared Floor scale reading: _____	
A6	MP-C.SLQ001	'Feed (remaining)' value on system software is within ± 0.2 kg of floor scale reading.	'Feed (Remaining)' value: _____ ☐ Within ± 0.2 kg of floor scale reading	
A7	MP-C.SLQ001	Manually run Pump 01 and Pump 02 until membrane modules and retentate lines are filled.	☐ Yes ☐ No	
A8	MP-C.SLQ001	Confirm no leaks are observed in all connections.	☐ No leaks observed	
A9	MP-C.SLQ001	Fill Concentrate Reactor until scale reads ~30kg.	Floor scale reading: _____	
A10	MP-C.SLQ001	Set parameters on main page of system software. - Starting Volume: 30L - DV Setpoint: 2.0 DV	☐ Parameters set	
A11	MP-C.SLQ001	Click START PROCESS button to fill membrane modules with water. After membrane modules are filled with	☐ Membrane modules filled with water	

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Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
		water, ensure Permeate Valve is set to vertical position and open the four clamps on permeate line.	☐ Permeate Valve set to vertical position ☐ Open four clamps on permeate line	
A12	MP-C.SLQ001	Click Stop button when 'Perm. Total' value reads 60L on system software.	'Perm. Total' value: _____	
A13	MP-C.SLQ001	Mount Pneumatic Drum Stirrer into holder and turn on.	☐ Yes ☐ No	
A14	MP-C.SLQ001	Drain liquid in diafiltration system/tubing into waste tank.	☐ Yes ☐ No	
A15	MP-C.SLQ001	Calculate Concentration Permeate Required Value: Concentration Permeate Required = [12 (g/L)-Average Suspension Sample Concentration (g/L)] / 12 (g/L) * Initial Suspension Volume (L) + 16 (L)	Concentration Permeate Required (L): _____	
A16	MP-C.SLQ001	Tare Concentrate Reactor Floor Scale.	☐ Floor scale tared	
A17	MP-C.SLQ001	Transfer Suspension from plastic drum using the peristaltic pump via the product line (C.SLQ003) into Concentrate Reactor	☐ Yes ☐ No	
A18	MP-C.SLQ001	Manually run Pump 01 and Pump 02 until "Perm. Total" value reaches Concentration Permeate Required Value. Note: No High Purity Water will be added to Concentrate Reactor and concentrate Reactor Floor Scale reading will gradually decrease	"Perm. Total" value: _____	
A19	MP-C.SLQ001	Tare suspension floor scale and place product tank onto floor scale.	☐ Yes ☐ No	
A20	MP-C.SLQ001	Record weight of product tank (C.SLQ003) without the lid.	Product tank weight: _____ kg	
A21	MP-C.SLQ001	Transfer Suspension to plastic drum using the peristaltic pump via the product line (C.SLQ003).	☐ Yes ☐ No	
A22	MP-C.SLQ001	Take sample of suspension in 50mL tubes at the beginning, middle, and end of the transfer. Label each tube Aliquot A, Aliquot B, and Aliquot C, respectively.	☐ Yes ☐ No	

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Step	Procedure	Step Description	Status / Results	Performed By: (Initial/Date)
A23	MP-C.SLQ001	Place clean containers below Pump 01 and Pump 02. Disconnect retentate lines from pumps and allow suspension to drain into containers. Pour contents of the containers into the product tank.	☐ Yes ☐ No	
A24	MP-C.SLQ001	Record the initial weight of the suspension. Assumption: The initial weight (in kg) of the suspension is equal to the weight of product tank subtracted from the weight of the suspension and product tank	Initial Suspension Weight: _____ kg	
A25	MP-C.SLQ001	Seal product tank.	☐ Yes ☐ No	
A26	MP-C.SLQ001	Create product label for Suspension (L.SLQ005) indicating lot number and expiration date of suspension. Apply label to product tank.	☐ Yes ☐ No	
A27	MP-C.SLQ001	Place product tank in quarantine pending QC analysis of suspension.	☐ Yes ☐ No	
A28	MP-C.SLQ001	<u>Prepare KTF-2000 System for storage</u> Fill concentrate reactor with ~30kg of high purity water and manually start running Pump 01 and Pump 02. Allow Membrane Modules 1 and 2 and Retentate Lines 1A, 1B, 2A, and 2B are filled. Allow pumps to run for an additional 10 minutes.	☐ Yes ☐ No	
A29	MP-C.SLQ001	Switch Permeate Line Valve to horizontal (closed) position and Waste Line Valve to vertical (open) position. Place Waste Line through peristaltic pump and into the concentrate reactor. Manually run the peristaltic pump to fully drain the contents of the concentrate reactor into the waste tank.	☐ Yes ☐ No	
A30	MP-C.SLQ001	Create Storage Solution. Combine 20L of high purity water + 1.5mL of bleach solution (S.SLQ007).	☐ Yes ☐ No	
A31	MP-C.SLQ001	Pour Storage Solution into Concentrate Reactor. Close 4 Permeate Line Clamps and switch Permeate Line Valve to horizontal (closed) position and Waste Line Valve to vertical (open) position.	☐ Yes ☐ No	

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A32	MP-C.SLQ001	Manually run Pump 01 and Pump 02 until Membrane Modules 1 and 2 and Retentate Lines 1A, 1B, 2A, and 2B are filled.	☐ Yes ☐ No	
A33	MP-C.SLQ001	Once filled, stop Pump 01 and Pump 02 and close the ball valves for Retentate Lines 1B and 2B at the Concentrate Reactor. Allow 5 minutes to elapse for settling and degassing. Ensure the Storage Solution has filled Membrane Modules 1 and 2 entirely. Storage Solution should at least fill Retentate Lines 1B and 2B up to the respective line's Flow Meter.	☐ Yes ☐ No	
A34	MP-C.SLQ001	Wait 5 minutes to ensure the Membrane Modules are full and stable.	☐ Yes ☐ No	
A35	MP-C.SLQ001	Manually run the Peristaltic Pump to fully drain the contents of the Concentrate Reactor into the Waste Tank.	☐ Yes ☐ No	
A36	MP-C.SLQ001	Power down Pump 03, Pump 04, concentrate reactor scale, suspension scale, and KTF system software.	☐ Yes ☐ No	
A37	MP-C.SLQ001	<u>Prepare the high purity water tank for storage</u> Take a peristaltic pump with sterilized tubing to pump out any residual high purity water from the tank.	☐ Yes ☐ No	
A38	MP-C.SLQ001	Allow high purity tank to dry for 24 hours before securing lid to the tank.	☐ Yes ☐ No	
A39	MP-C.SLQ001	Clean containers using in the manufacturing process. Allow containers to dry and store after they have dried.	☐ Yes ☐ No	
A40	QC-C.SLQ001	Execution of QC analysis of suspension samples. QC analysis report attached to DHR.	☐ Yes ☐ No	
A41	MP-C.SLQ001	Attach Suspension Quality Control Report and COA to DHR.	☐ Yes ☐ No	
A42	MP-C.SLQ001	Test results meet acceptance criteria.	☐ Yes ☐ No	
A43	MP-C.SLQ001	<u>Dilution Procedure</u> Calculate the Average Suspension Sample Concentration (g/L) by taking the average 'Residue Net Weight' of six (6) freeze	Average Suspension Sample Concentration: _____ g/L	

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<u>Step</u>	<u>Procedure</u>	<u>Step Description</u>	<u>Status / Results</u>	<u>Performed By:</u> <u>(Initial/Date)</u>
		<u>dried suspension samples based off the 10 mL sample.</u>		
<u>A44</u>	<u>MP-C.SLQ001</u>	<u>Determine if Concentration Process is required. If Average Suspension Sample Concentration is <10g/L, Concentrate Process is required again. If Average Suspension Sample Concentration is >10g/L, Dilution Procedure can proceed.</u>	<u>Is Concentration Process Required?</u> ☐ Yes, skip Steps A45-A50 and proceed to Step A1 and repeat. ☐ No, proceed to Step A45.	
<u>A45</u>	<u>MP-C.SLQ001</u>	<u>Calculate the volume (L) of high purity water to be added to the product tank.</u> <u>Volume = [Average Suspension Sample Concentration (g/L) * Initial Suspension Volume (L)]/[10 (g/L)] - Initial Suspension Volume (L)</u>	<u>Volume of high purity water to be added to product tank:</u> _____ L	
<u>A46</u>	<u>MP-C.SLQ001</u>	<u>Tare suspension floor scale and place product tank onto floor scale.</u>	☐ Yes ☐ No	
<u>A47</u>	<u>MP-C.SLQ001</u>	<u>Add volume of high purity water (from Step 35) to the product tank.</u>	☐ Yes ☐ No	
<u>A48</u>	<u>MP-C.SLQ001</u>	<u>Record the weight of product tank with suspension</u>	_____ kg	
<u>A49</u>	<u>MP-C.SLQ001</u>	<u>Calculate weight of suspension.</u>	_____ kg	
<u>A50</u>	<u>MP-C.SLQ001</u>	<u>Update product label with weight of suspension.</u>	☐ Yes ☐ No	

End of procedure

Comments/Additional Notes:

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Suspension Analysis

Quality Control Procedure



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1 Purpose

- 1.1 The purpose of this quality control procedure is to outline the steps to determine the quality attributes and yield of the manufactured suspension C.SLQ001.

2 Scope

- 2.1 This procedure is applies to the final quality control step of Silq's Suspension manufactured at Silq's manufacturing facility.

3 Documents

- 3.1 Reference Documents
- 3.1.1 MP-C.SLQ001 Manufacturing Procedure, Suspension Processing
 - 3.1.2 SP-C.SLQ001 Specification, Suspension
 - 3.1.3 QM.SLQ049 Workstation Practices SOP
- 3.2 Associated Documents / Forms
- 3.2.1 TR-C.SLQ001 Suspension Processing Traveler
 - 3.2.2 TMP1-QC-C.SLQ001 Template, Suspension Quality Control Report
 - 3.2.3 TMP2-QC-C.SLQ001 Template, Certificate of Analysis, Suspension

4 Definitions

- 4.1 Abbreviations / Acronyms
- 4.1.1 **COA** – Certificate of Analysis
 - 4.1.2 **DHR** – Device History Record
 - 4.1.3 **F** – Fluorine
 - 4.1.4 **H** – Hydrogen
 - 4.1.5 **mL** – Milliliter
 - 4.1.6 **NMR** – Nuclear Magnetic Resonance
 - 4.1.7 **pH** – Potential Hydrogen
 - 4.1.8 **ppm** – Parts per Million
 - 4.1.9 **QA** – Quality Assurance
 - 4.1.10 **TFE** – Trifluoroethanol
- 4.2 General Definitions



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4.2.1 N/A

5 Responsibilities

- 5.1 It is the responsibility of Manufacturing/Operations management or their authorized representative to ensure this procedure is adequate and appropriate for fulfilling the purpose of this document.
- 5.2 It is the responsibility of Silq personnel and approved contractor personnel to perform their job duties in compliance with the requirements of this procedure and its associated procedures, as appropriate, including ensuring that any output documentation from this procedure is complete and accurate.

6 Safety Information

- 6.1 Gloves (latex or nitrile) and safety eye wear are to be worn at all times during the execution of this procedure.

7 Equipment / Tools / Fixtures

- 7.1 Weighing scale (E.SLQ013)

8 Components / Materials / Supplies

- 8.1 High purity water (S.SLQ001)
- 8.2 Centrifuge tube, 50 mL, Polypropylene, Flat Cap (S.SLQ002)
- 8.3 Container, 8.5" x 18" x 6" (S.SLQ003)
- 8.4 Syringe, 10mL (S.SLQ004)
- 8.5 pH Strip (S.SLQ005)
- 8.6 Deuterium oxide (S.SLQ006)
- 8.7 Permanent pen

9 Procedure: Line Clearance

- 9.1 Perform line clearance of the work area in accordance with Workstation Practices QM.SLQ049.

10 Procedure: General Requirements

- 10.1 This quality control procedure will be performed and the results of the procedure shall be documented using the Suspension Quality Control Report template TMP1-QC-C.SLQ001.



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- 10.2 Results of the quality control test shall be documented on the Certificate of Analysis template TMP2-QC-C.SLQ001 and will determine if the batch/lot of Suspension manufactured meets established acceptance criteria.

11 Procedure: Sample Preparation

- 11.1 Obtain the Suspension samples in the three (3) 50mL centrifuge tubes labeled Aliquot A, Aliquot B, and Aliquot C.
- 11.2 Label six (6) empty 50mL centrifuge tubes (S.SLQ002): Sample A1, Sample A2, Sample B1, Sample B2, Sample C1, and Sample C2.
- 11.3 Weigh the empty tubes and record the weight of the tubes on the Suspension Quality Control Report.
- 11.4 From each Aliquot tube, extract two (2) 10mL aliquots using a 10mL syringe and place each 10mL aliquot in their respective sample tubes. For example, two (2) 10mL aliquots from Aliquot A suspension sample tube will go into Sample A1 tube and Sample A2 tube, respectively. Do the same for Aliquot B and Aliquot C suspension samples.
- 11.4.1 The remaining contents of each Suspension sample tube (Aliquot A, Aliquot B, and Aliquot C) will be stored in reserve if further analysis is needed.

- 11.5 Freeze dry the sample tubes. After freeze drying, weigh each sample tube and record the weight ('Weight with Residue') in the Suspension Quality Control Report.

11.5.1 Using the aliquot volume of 10mL (0.01L), calculate the Suspension Sample Concentration for each tube and record in the Suspension Quality Control Report.

11.5.2 Calculate the Average Suspension Sample Concentration by calculating the average Suspension Sample Concentration from each tube and record in the Suspension Quality Control Report and the Suspension Processing Traveler.

- 11.6 Weigh an empty 50mL tube using the weighing scale (E.SLQ013) and label it 'Suspension Test Sample.' Record empty weight in the Suspension Quality Control Report. The dried contents of sample tubes A1, B1, and C1 will be combined into the 'Suspension Test Sample' tube and then resuspended with High Purity Water (S.SLQ001) to a concentration of 1% w/v and subject to analysis of Appearance (color), Appearance (form), Residue on Evaporation, Specific Gravity, and pH.

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- 11.7 The dried contents of sample tubes A2, B2, and C2 will be combined into a separate 50mL centrifuge tube labeled 'NMR Test Sample' and then resuspended in 5mL of deuterium oxide (D₂O) (S.SLQ006). A Bruker AV400 broad band FT NMR Spectrometer is used to obtain ¹H and ¹⁹F NMR Spectra. The Spectrometer is managed by the UCLA Molecular Instrumentation Center (MIC), a state-of-the-art campus-wide facility dedicated to enabling the use of modern instrumentation in molecular characterizations (www.mic.ucla.edu). The MIC is housed within and managed through the UCLA Department of Chemistry and Biochemistry. The professional staff consists of seven Ph.D. level staff members and graduate student assistants. The staff is responsible for operating, maintaining, and upgrading the equipment, training users, and providing consultation for the application of modern instrumentation.

12 Quality Control Test Procedure

12.1 Appearance

- 12.1.1 Visually inspect the Suspension test sample for color appearance. Record result in the Suspension Quality Control Report.

12.1.2 Acceptance criteria:

12.1.2.1 Color: Opaque white

12.1.2.2 Form: Inviscid liquid

12.2 Residue on Evaporation

- 12.2.1 Obtain a 50mL tube and label it 'Residue on Evaporation Sample'.

- 12.2.2 Weigh the empty 50mL tube and record the weight in the Suspension Quality Control Report.

- 12.2.3 Obtain a 10mL aliquot from the Suspension Test Sample tube using a 10mL syringe and place into the 'Residue on Evaporation Sample' tube. Weight the 'Residue on Evaporation Sample' tube and record value.

- 12.2.4 Freeze dry the 'Residue on Evaporation Sample' tube for 48 hours so that all water is removed from the tube.

- 12.2.5 Weigh the 'Residue on Evaporation Sample' tube and record the weight ('Weight with Residue on Evaporation') in the Suspension Quality Control Report. Calculate the Residue on Evaporation %.

Residue on Evaporation % = (fill in)

12.2.6 Acceptance criteria:

12.2.6.1 Residue on Evaporation: 0.5% – 1.5%

12.3 pH



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12.3.1 Fully insert a pH strip (S.SLQ005) into the Suspension test sample tube for 5 seconds and remove it. Allow the test strip to dry for 10 seconds.

12.3.2 Match the resulting color of the test strip to a pH level using the provided key. Record the pH value in the Suspension Quality Control Report.

12.3.3 Acceptance criteria:

12.3.3.1 5 – 8

12.4 ¹H NMR

12.4.1 Send the Obtain the 'NMR Test Sample' prepared in Section 11.7 to the approved test laboratory and obtain the ¹H NMR spectra.

12.4.2 Attach the spectra to the Suspension Quality Control Report.

12.4.3 Acceptance criteria

12.4.3.1 A successful result is one that includes broad peaks within 0.2ppm of 0.99, 2.2, 2.9, 3.1, 3.5, 3.7, 4.4, and 4.75 ppm indicating successful incorporation of the zwitterion moiety into the polymer. The peak at 4.75 ppm is the 'solvent peak' resulting from the deuterium oxide solvent. All peaks should be easily identifiable and have an amplitude of at least 5% of the solvent peak. An acceptable ¹H NMR spectra plot is shown in Appendix A.

12.5 ¹⁹F NMR

12.5.1 Send the Obtain the 'NMR Test Sample' prepared in Section 11.7 to the approved test laboratory and obtain the ¹⁹F NMR spectra.

12.5.2 Attach the spectra to the Suspension Quality Control Report.

12.5.3 Acceptance criteria

12.5.3.1 A successful result is one that includes broad peaks within 5ppm of -140 and -150ppm indicating successful incorporation of the PFPA moiety into the polymer. These peaks should be broad with a small amplitude but still easily identifiable. If TFE is present in the polymer due to an incomplete diafiltration process during manufacturing, a large, sharp peak will also appear at -76ppm. If this peak is greater than 4X the amplitude of the peaks at -140 and -150ppm then the diafiltration process may be extended until a successful ¹⁹F NMR spectra is obtained. An acceptable ¹⁹F NMR spectra plot is shown in Appendix B. An unacceptable ¹⁹F NMR spectra plot is shown in Appendix C.



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12.6 Sign and date the completed Suspension Quality Control Report and include the report in the batch DHR.

13 Procedure: Certificate of Analysis

13.1 Create a certificate of analysis using the COA template TMP2-QC-C.SLQ001.

13.1.1 Enter the test results for the applicable attributes based on the recorded results in the Suspension Quality Control Report.

13.1.2 Print out the COA. Sign and date the COA.

13.1.3 Attach a copy of the test results and COA to the batch DHR.

13.1.4 Include the original, signed COA with the product tank.



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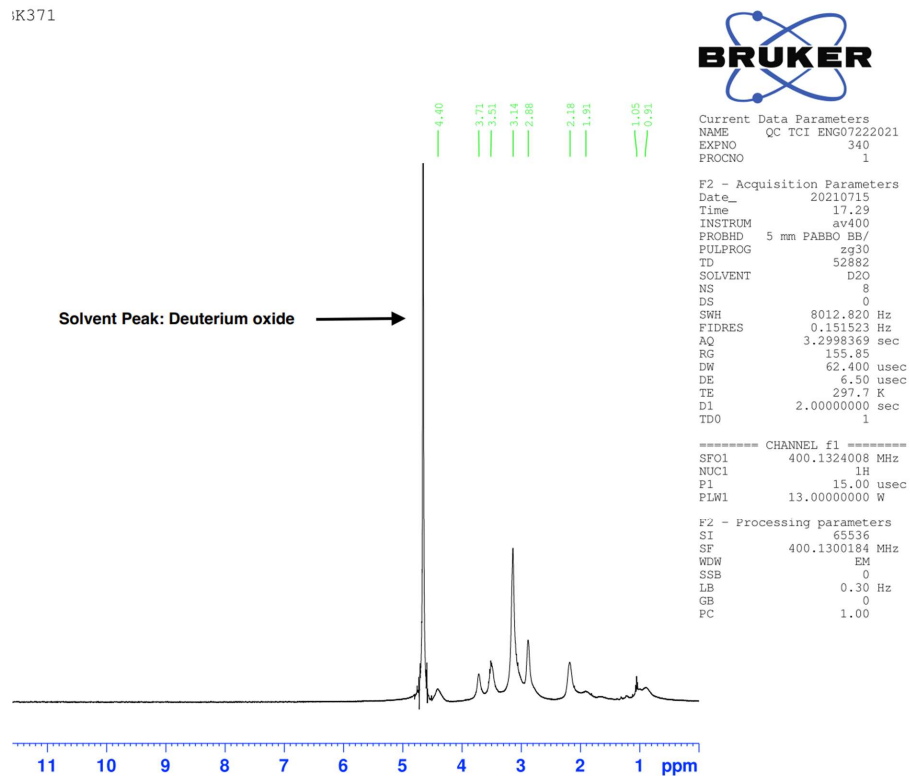
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14 Appendix A: ¹H NMR Spectra (Acceptable)

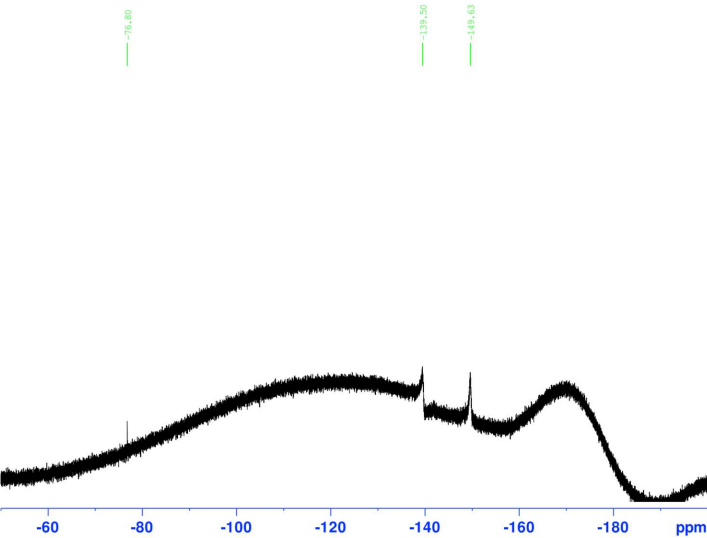
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15 Appendix B: ¹⁹F NMR Spectra (Acceptable)

Account No. RBK371
PSB2



Current Data Parameters
NAME QC TCI CoFA Lot_08252021
EXPNO 391
PROCNO 1

F2 - Acquisition Parameters
Date_ 20210824
Time 19.41
INSTRUM av400
PROBHD 5 mm PABBO BB/
PULPROG zgpg30p2
TD 62144
SOLVENT D2O
NS 256
DS 0
SWH 150000.000 Hz
FIDRES 0.572205 Hz
AQ 0.8738133 sec
RG 189.85
DM 3.333 usec
DE 6.50 usec
TE 297.5 K
D1 1.00000000 sec
D11 0.03000000 sec
D12 0.00002000 sec
TD0 1

===== CHANNEL F1 =====
SFO1 376.4607162 MHz
NUC1 19F
P1 14.50 usec
PLM1 17.00000000 W

===== CHANNEL F2 =====
SFO2 400.1324008 MHz
NUC2 1H
CPDPRG2 waltz16
PCPD2 30.00 usec
PLM2 13.00000000 W
PLM22 0.3611000 W

F2 - Processing parameters
SI 262144
SF 376.4983660 MHz
WDW EM
SSB 0
LB 1.00 Hz
GB 0
PC 1.00



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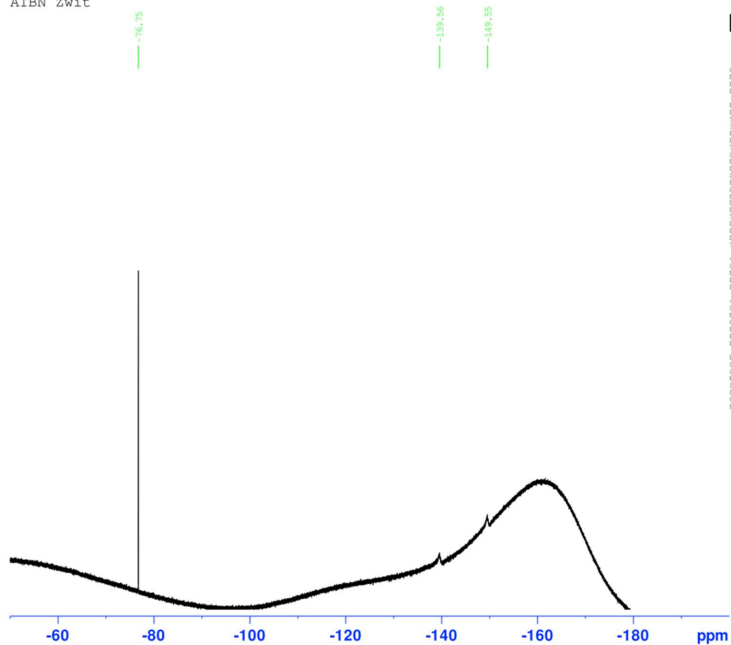
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16 Appendix C: ^{19}F NMR Spectra (Not Acceptable)

Account No. RBK371
AIBN Zwit



BRUKER

Current Data Parameters
NAME PDB-PFPA 1H,19F Dialysis Bottle 4day
EXPNO 1
PROCNO 1

F2 - Acquisition Parameters
Date_ 20180709
Time 13.33
INSTRUM spect
PROBHD 5 mm PABBO 1H/1
PULPROG zgpg30
TD 65536
SOLVENT DMS
NS 800
DS 4
SWH 18000.000 MHz
FIDRES 0.470200 Hz
AQ 0.4718133 sec
RG 384.0
SQ 7.133 sec
DE 45.00 dB
TE 300.2 K
D1 1.0000000 sec
d11 0.0300000 sec
d12 0.0002000 sec
TD0 1

===== CHANNEL f1 =====
NUC1 19F
DECT 129F
F1 498.50 MHz
P1 17.00000000 W

===== CHANNEL f2 =====
NUC2 1H
DECT 1H
F2 500.136000 MHz
P2 13.00000000 W
SFO2 500.136000 MHz

F2 - Processing parameters
SI 32768
SF 500.136000 MHz
WDW EM
SSB 0
LB 1.00 Hz
GB 0
PC 1.00